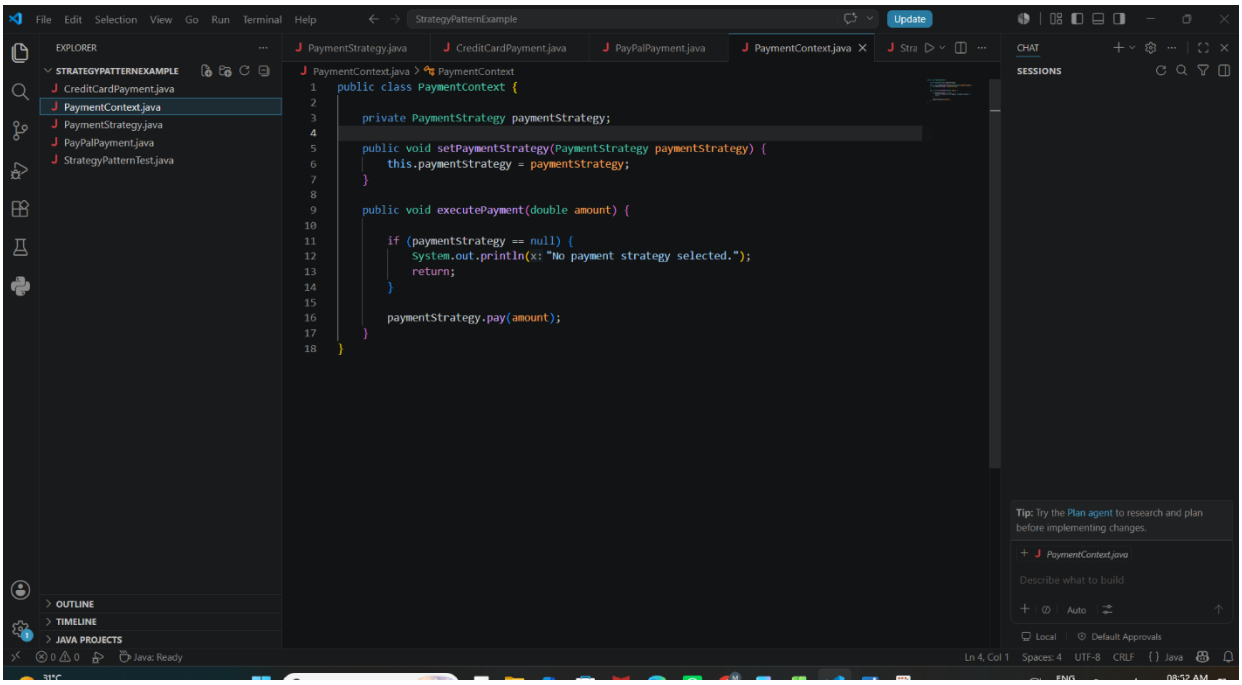
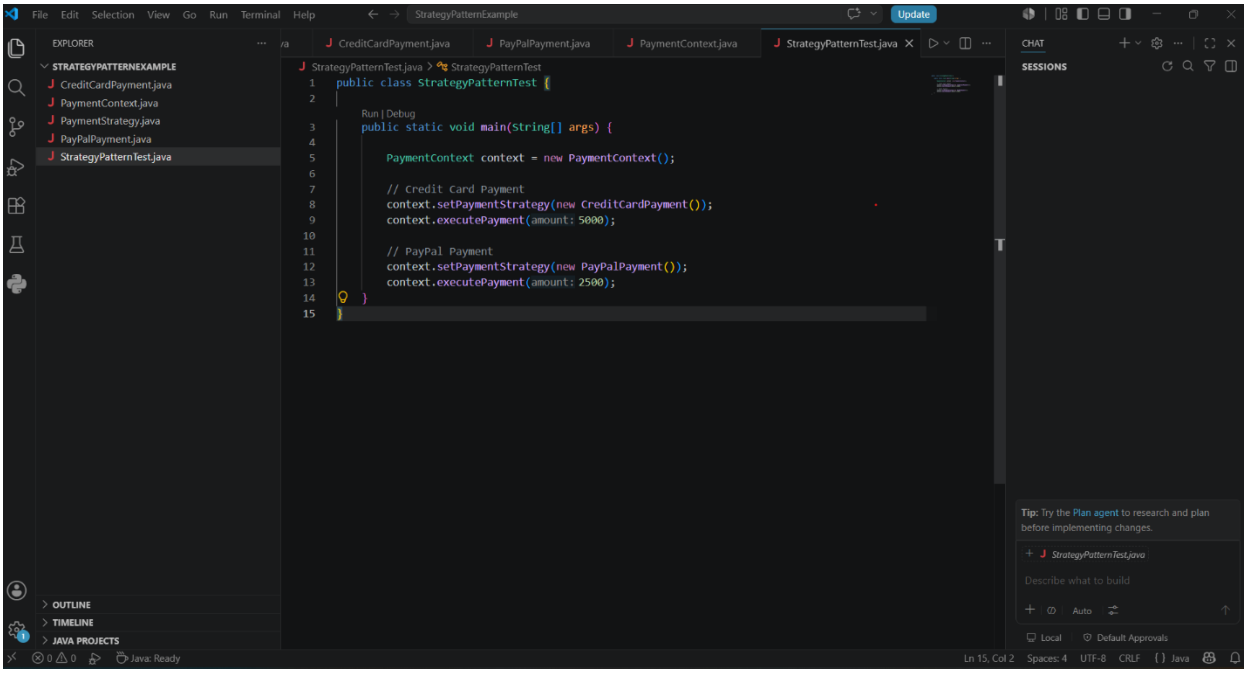
## Hands-On

### Exercise 8: Implementing the Strategy Pattern

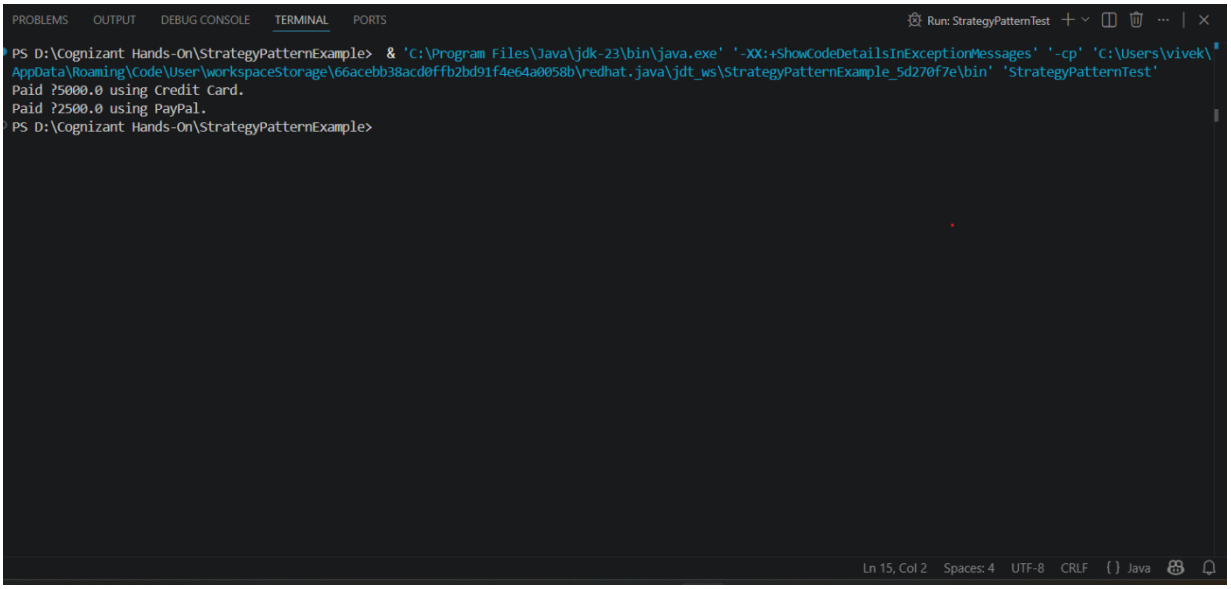
As part of this module, I successfully implemented, executed, and tested the Strategy Pattern in Java by developing a project named StrategyPatternExample. I created a PaymentStrategy interface with a pay() method and implemented concrete strategy classes such as CreditCardPayment and PayPalPayment, each providing its own payment processing logic. I also developed a PaymentContext class that maintained a reference to a PaymentStrategy object and executed the selected payment strategy at runtime. After completing the implementation, I ran the program and tested different payment methods by dynamically switching between strategies to verify their functionality. This hands-on exercise helped me understand how the Strategy Pattern enables the selection of algorithms or behaviors at runtime, promotes flexibility, reduces conditional logic, and makes applications easier to extend and maintain.

### Coding:




File Edit Selection View Go Run Terminal Help
StrategyPatternExample
Update
EXPLORER
STRATEGYPATTERNEXAMPLE
CreditCardPayment.java
PaymentContext.java
PaymentStrategy.java
PayPalPayment.java
StrategyPatternTest.java
CreditCardPayment.java
PayPalPayment.java
PaymentContext.java
StrategyPatternTest.java
StrategyPatternTest.java > StrategyPatternTest
1 public class StrategyPatternTest {
2
Run | Debug
3 public static void main(String[] args) {
4
5 PaymentContext context = new PaymentContext();
6
7 // Credit Card Payment
8 context.setPaymentStrategy(new CreditCardPayment());
9 context.executePayment(amount: 5000);
10
11 // PayPal Payment
12 context.setPaymentStrategy(new PayPalPayment());
13 context.executePayment(amount: 2500);
14 }
15 }
CHAT
SESSIONS
Tip: Try the Plan agent to research and plan before implementing changes.
StrategyPatternTest.java
Describe what to build
Auto
Local
Default Approvals
OUTLINE
TIMELINE
JAVA PROJECTS
Java: Ready
Ln 15, Col 2
Spaces: 4
UTF-8
CRLF
Java

# Output:


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Run: StrategyPatternTest
PS D:\Cognizant Hands-On\StrategyPatternExample> & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\vivek\AppData\Roaming\Code\User\workspaceStorage\66acebb38acd0ffb2bd91f4e64a0058b\redhat.java\jdt_ws\StrategyPatternExample_5d270f7e\bin' 'StrategyPatternTest'
Paid ?5000.0 using Credit Card.
Paid ?2500.0 using PayPal.
PS D:\Cognizant Hands-On\StrategyPatternExample>
Ln 15, Col 2
Spaces: 4
UTF-8
CRLF
Java